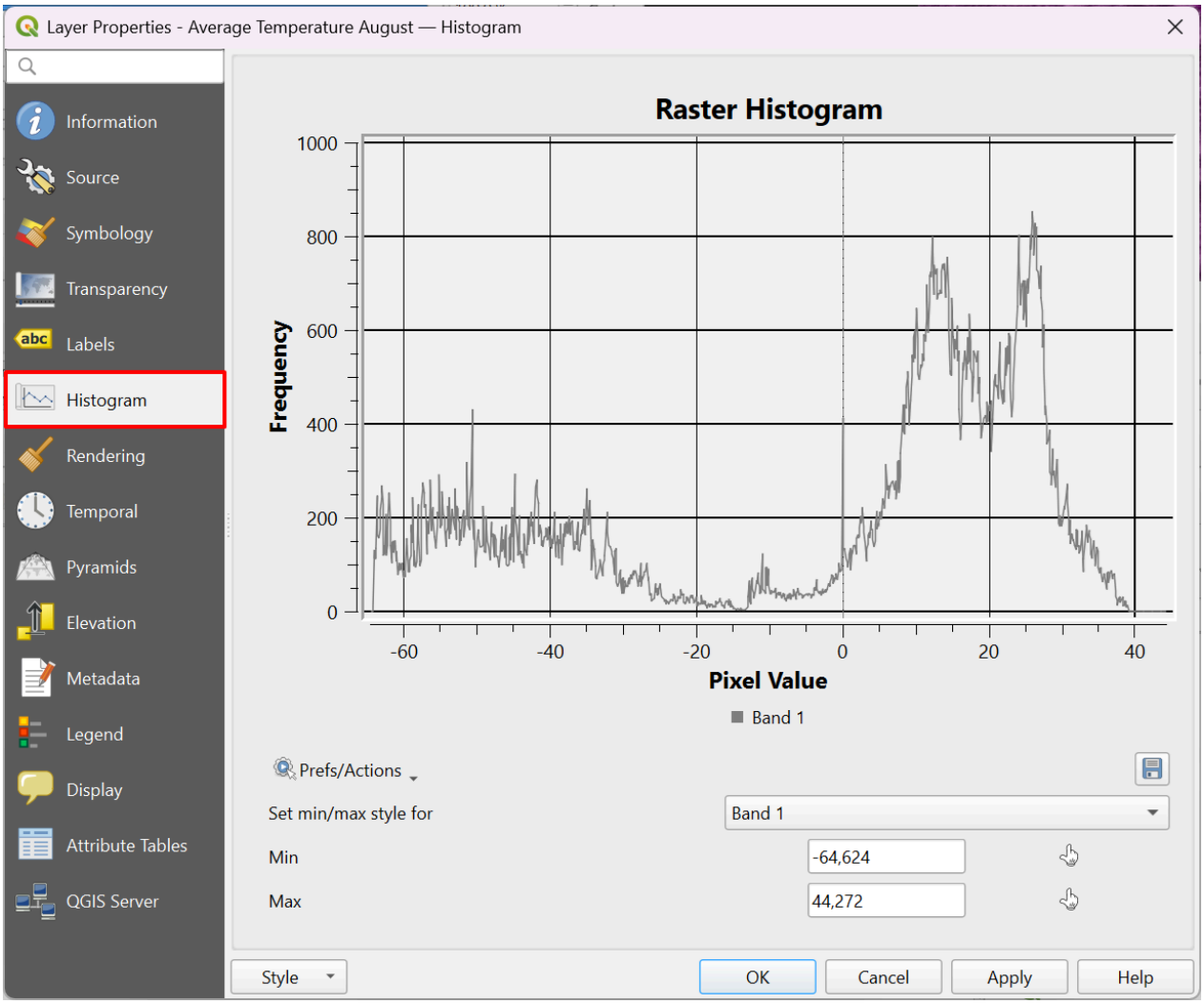
# 4.3.1 Cell values: histogram and unique values

Each cell in a band of a raster layer has one numeric value or *NoData* (see 2.1 The raster model). A raster usually contains thousands or even millions of cells, and a cell rarely stands for a clearly delimited object, such as a building or a parcel. A list of all cells, like the attribute table of a vector layer, would therefore contain an unmanageable number of rows and tell you little. What is useful is an overview of **how often each value occurs**. You can get such an overview in QGIS in several ways. Below we discuss two of them: a histogram and a report of unique values.

> Warning
>
> These overviews count the values **per image band**: in a multi-band raster, each band has its own values. The histogram shows a separate curve for each band, and the report always covers one band (see Single-band and multi-band).

## Histogram of cell values

Open the *Layer Properties* window of your raster layer and go to the *Histogram* tab. If the histogram is still empty, you can compute it by clicking *Compute Histogram*. The x-axis shows the cell values (*Pixel Value*), and the y-axis how often each value occurs (*Frequency*).



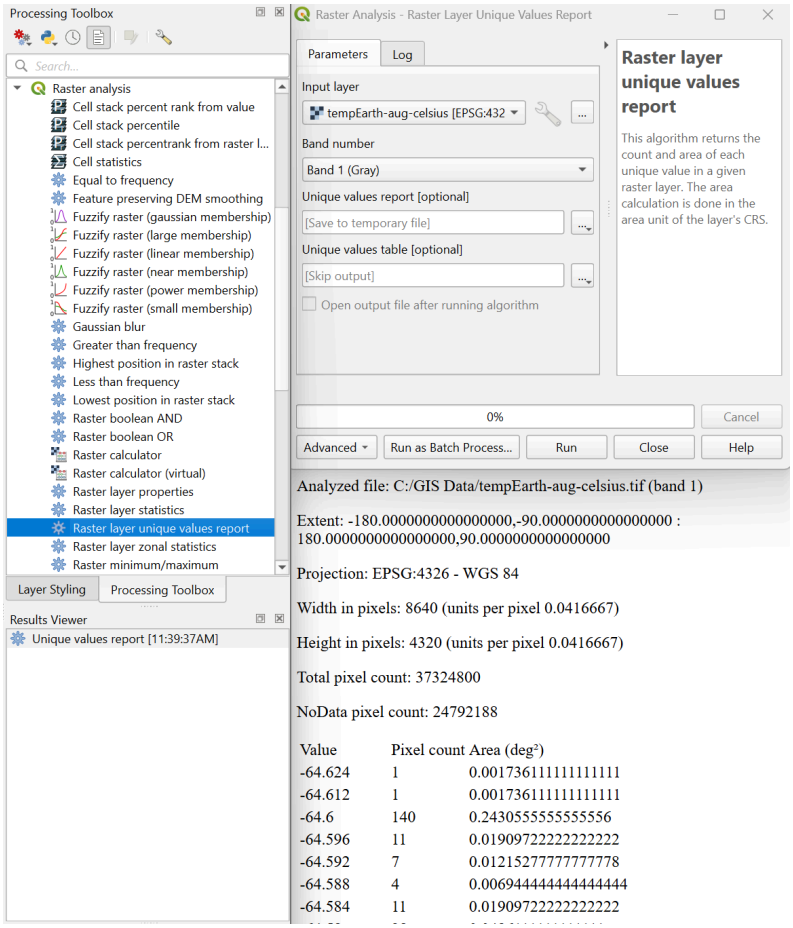

Tip

If you drag the mouse over an area of the histogram, QGIS zooms in on that area. To return to the full view, press *Esc*.

## Report of unique cell values

A second way to get an overview is a **report** that lists every unique value of a band, with the number of cells that have it. You create this report with an analysis tool. QGIS gathers all its processing and analysis tools in the *Processing Toolbox*, which by default is on the right side of the QGIS window. If it is not visible, open it from the main menu with *Processing* ▸ *Toolbox*.

Under *Raster analysis*, double-click the tool *Raster layer unique values report*. In the window that opens, choose the layer (*Input layer*) and the band (*Band number*), and click *Run*. When the report is ready, QGIS opens the *Results Viewer* panel with the report in it: click the report to open it. Besides the unique values, the report also gives the number of NoData cells.




Analyzed file: C:/GIS Data/tempEarth-aug-celsius.tif (band 1)

Extent: -180.0000000000000000,-90.0000000000000000 : 180.0000000000000000,90.0000000000000000

Projection: EPSG:4326 - WGS 84

Width in pixels: 8640 (units per pixel 0.0416667)

Height in pixels: 4320 (units per pixel 0.0416667)

Total pixel count: 37324800

NoData pixel count: 24792188

| Value | Pixel count | Area (deg²) |
|---|---|---|
| -64.624 | 1 | 0.0017361111111111111 |
| -64.612 | 1 | 0.0017361111111111111 |
| -64.6 | 140 | 0.2430555555555556 |
| -64.596 | 11 | 0.01909722222222222 |
| -64.592 | 7 | 0.01215277777777778 |
| -64.588 | 4 | 0.006944444444444444 |
| -64.584 | 11 | 0.01909722222222222 |